

14. | Write a code using RAW sockets to implement packet sniffing.

**AIM:** To write a Python program using RAW sockets to implement packet sniffing, capturing IP packets and displaying their source IP address and packet size.

**packet\_sniffer.py**

```
import socket

HOST = socket.gethostname(socket.gethostname())

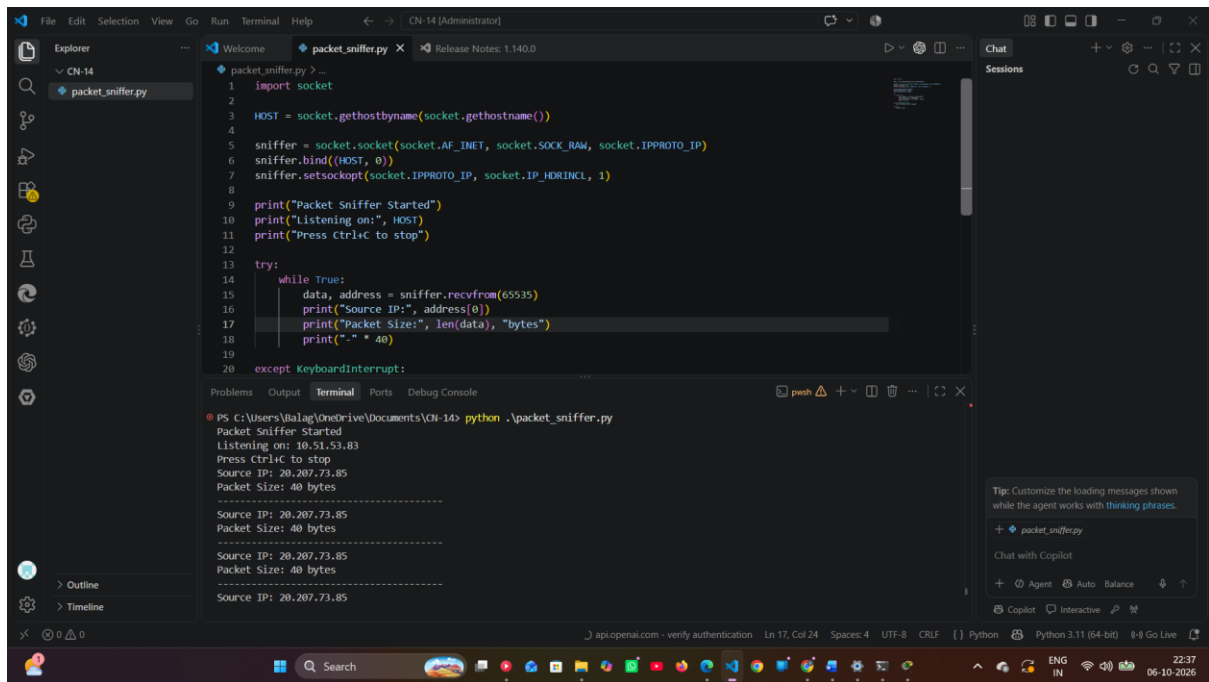
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
sniffer.bind((HOST, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

print("Packet Sniffer Started")
print("Listening on:", HOST)
print("Press Ctrl+C to stop")

try:
    while True:
        data, address = sniffer.recvfrom(65535)
        print("Source IP:", address[0])
        print("Packet Size:", len(data), "bytes")
        print("-" * 40)

except KeyboardInterrupt:
    print("\nPacket Sniffer Stopped")

finally:
    sniffer.close()
```



The image shows a Visual Studio Code editor window with a Python script named `packet_sniffer.py` open. The script is a packet sniffer that listens on a specific IP address and port, receiving and displaying incoming network packets. The terminal output shows the script running successfully, displaying the source IP and packet size for several received packets.

```
1 import socket
2
3 HOST = socket.gethostname(socket.gethostname())
4
5 sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
6 sniffer.bind((HOST, 0))
7 sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
8
9 print("Packet Sniffer Started")
10 print("Listening on:", HOST)
11 print("Press Ctrl+C to stop")
12
13
14 try:
15     while True:
16         data, address = sniffer.recvfrom(65535)
17         print("Source IP:", address[0])
18         print("Packet Size:", len(data), "bytes")
19         print("-" * 40)
20 except KeyboardInterrupt:
```

Terminal Output:

```
PS C:\Users\Balag\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
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Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
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-----
```

The screenshot shows a Windows 11 desktop environment. In the foreground, a Visual Studio Code (VS Code) window is open, displaying a terminal session. The terminal title bar indicates the command prompt is running as 'CN-14 [Administrator]'. The terminal output shows the execution of a Python script named 'packet\_sniffer.py'. The script starts by listening on the IP address 10.51.53.83. It then captures several network packets, displaying their source IP addresses and sizes. The packets are separated by dashed lines for readability. The source IP addresses include 20.207.73.85, 10.51.53.38, 104.18.39.21, 150.171.110.86, and 140.82.113.21. The packet sizes are mostly 40 bytes, except for one which is 289 bytes. The terminal session concludes with the message 'Packet Sniffer Stopped'. In the background, the Windows taskbar is visible, showing the Start button, a search bar, and several pinned application icons including File Explorer, Microsoft Edge, and WhatsApp. The system tray at the bottom right shows the date and time as 10:00 AM on 10/10/2023.

```
PS C:\Users\Balag\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet Sniffer Started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 10.51.53.38
Packet Size: 289 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Packet Sniffer Stopped
```